

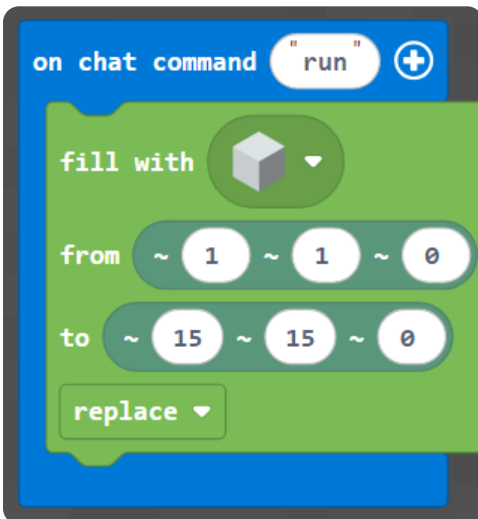
3D Coordinates Extension — English Worksheet (Beginner)

Topic: Design Your Own Pixel Art (x, y) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Lesson 3) · **Time:** about 30 minutes

Every other week the picture was given to you. This lesson **you** draw it — then **you** read your own drawing back onto the wall.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🌟

Read the steps. Circle your answer.

```
place blue block at (3, 9)
place blue block at (4, 9)
place blue block at (5, 9)
```

A line across or up? Circle one: across · up

Which number stays the same? Circle one: x · y

```
place blue block at (7, 2)
place blue block at (7, 3)
place blue block at (7, 4)
```

Across or up? Circle one: across · up

Which number changes? Circle one: $x \cdot y$

2 · Spot the Bug 🐛

This code is broken. Read what it should do, then fill in the fix.

Bug — The flower's two leaves should sit at **two different spots**: (2, 4) and (4, 4). Both blocks land in the **same spot**.

```
place green block at (2, 4)
place green block at (2, 4)
```

Hint: the first number (x) of the second leaf is wrong. One leaf is further **across**, so its x must be **bigger**.

Fill the blank to fix the second line:

```
place green block at (2, 4)
place green block at (__, 4)
```

Which number fills the blank? Circle one: 2 · 4

Why was the code wrong? Circle one: both leaves had the same x · the colour was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Your five rules.

#	Your design must...
1	fit inside the 15 × 15 wall
2	use 3 or more different block colours
3	use 25 or more blocks
4	be recognisable — someone else can name it without being told
5	be drawn on the grid below before you place a single block

Here is one I made — 3 colours, and you can tell what it is:

🌸 **Flower**

	1	2	3	4	5
5			Red		
4		Red	Red	Red	
3	Red	Red	Black	Red	Red
2		Red	Red	Red	
1			Green		

Now draw yours. Colour the squares. x goes **across**, y goes **up**.

Your design

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15															
14															
13															
12															
11															
10															
9															
8															
7															
6															
5															
4															
3															
2															
1															

Then build it. Read each coloured square's (x, y) and place that block on your wall.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your grid. Hold this paper up to the camera.

2 • Your build. Point the camera. Hold your paper next to the wall so both are in the shot.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.